[728] (CONCLUDED)
A THEOREM IN ELLIPTIC FUNCTIONS.

that is,

$$(x'^2 - v'^2)(1 - k^2 v'^2 v''^2)(u^2 - y'^2)(1 - k^2 u^2 y'^2)$$
$$= 2\big[xw\sqrt{1-y'^2}\sqrt{1-k^2y'^2}\sqrt{1-v''^2}\sqrt{1-k^2v''^2} + yz\sqrt{1-x'^2}\sqrt{1-k^2x'^2}\sqrt{1-v'^2}\sqrt{1-k^2v'^2}\big]^2.$$

Rationalising, we obtain, as mentioned above, an equation containing only the squares x^2, y^2, z^2, w^2; it therefore is of a degree twice that of the equation containing the product $xyzw$. I worked out in this way the equation in (x^2, y^2, z^2, w^2), but the calculation was lost, and the easier way of obtaining it is obviously by means of the equation involving $xyzw$.

We have, by the theorem,

$$\frac{1 - k^2 \cdot xyzw}{\sqrt{1-x^2}\sqrt{1-k^2x^2}\sqrt{1-y^2}\sqrt{1-k^2y^2}\sqrt{1-z^2}\sqrt{1-k^2z^2}\sqrt{1-w^2}\sqrt{1-k^2w^2}} = \frac{1}{k^2};$$

that is,

$$k^2(1-k^2xyzw) = k^2\sqrt{1-x^2}\sqrt{1-k^2x^2}\sqrt{1-y^2}\sqrt{1-k^2y^2}\sqrt{1-z^2}\sqrt{1-k^2z^2}\sqrt{1-w^2}\sqrt{1-k^2w^2},$$

and then, writing

$$\begin{aligned}
P &= x^2+y^2+z^2+w^2,\\
Q &= x^2y^2+x^2z^2+x^2w^2+y^2z^2+y^2w^2+z^2w^2,\\
R &= x^2y^2z^2+x^2y^2w^2+x^2z^2w^2+y^2z^2w^2,\\
S &= x^2y^2z^2w^2,
\end{aligned}$$

and using $\sqrt{S}$ to denote the rational function $xyzw$, we have

$$\begin{aligned}
&k^2(1-2k^2\sqrt{S}+k^4S)\\
&= k^2(1-P+Q-R+S)\\
&-2k^2\sqrt{(1-P+Q-R+S)(1-k^2P+k^4Q-k^6R+k^8S)};
\end{aligned}$$

or, if for a moment the radical is called $\sqrt{\Delta}$, then the factor k^2 divides out, and the equation becomes

$$\sqrt{\Delta} = \tfrac{P}{2} - \tfrac{Q}{2} + R - \tfrac{S}{2} + k^2(-\tfrac{P}{2} + Q - R + \tfrac{S}{2}) + \cdots,$$

whence

$$\begin{aligned}
&4(1-P+Q-R+S)(1-k^2P+k^4Q-k^6R+k^8S)\\
&= \big\{P-2+(1+k^2)P-2k^2Q-(k^2+k^4)R+2k^4S\big\}^2 - 4k^4S\\
&= \big\{-2+P+k^2P+\cdots - 2k^4\sqrt{S}\cdot\{2-(1+k^2)P+2k^2Q-(k^2+k^4)R+2k^4S\}\big\}.
\end{aligned}$$

The factor k^4 divides out; omitting it, we have

$$\begin{aligned}
&4Q - P^2 - 4(1+k^2)R + 16k^2S + 2k^2PR - 2(k^2+k^4)PS - k^4R^2 + 4k^4QS\\
&= -2\sqrt{S}\{2-(1+k^2)P+2k^2Q-(k^2+k^4)R+2k^4S\},
\end{aligned}$$

or, as this may also be written,

$$\begin{aligned}
&\{(-P^2+4Q-4R)+k^2(-4P+2PR+16S-4PS)+k^4(-R^2+4QS-PS)\}\\
&= -2\sqrt{S}\,\{2-P+k^2(-P+2Q-R)+k^4(-R+2S)\},
\end{aligned}$$

which is the required rational equation involving the product of the sn's.

729.

ON A THEOREM RELATING TO CONFORMABLE FIGURES.

[From the *Proceedings of the London Mathematical Society*, vol. x. (1879), pp. 143–146.
Read May 8, 1879.]

. Consider two plane figures, say the figure of the points P referred to axes Ox, Oy, and that of the points P' referred to axes Ox', Oy'; and let x, y be the coordinates of P, and x', y' those of P'. If the figures correspond to each other in any manner whatever, P and P' being corresponding points, then we have x', y' each of them a function of x, y; and we may consider the second figure as derived from the first by altering the distance OP in the ratio $\sqrt{dx'^2+dy'^2} : \sqrt{dx^2+dy^2}$, and by rotating it through the angle $\tan^{-1}\dfrac{dy'}{dx'} - \tan^{-1}\dfrac{dy}{dx}$; say the Extension- and the Rotation-ratios.

and by the Rotation $\tan^{-1}\dfrac{dy'}{dx'} - \tan^{-1}\dfrac{dy}{dx}$, where the Extension and the Rotation are each of them a determinate function of x, y, the coordinates of P.

Passing from the point P to a consecutive point Q, the coordinates of which are $x+dx$, $y+dy$ (the ratio $dy : dx$ being arbitrary), then the coordinates of the corresponding point Q' will be $x'+dx'$, $y'+dy'$, where

$$dx' = \frac{dx'}{dx}dx + \frac{dx'}{dy}dy, \qquad dy' = \frac{dy'}{dx}dx + \frac{dy'}{dy}dy.$$

Writing $\dfrac{dx'}{dx}$ and $\dfrac{dy'}{dx}$ instead of $dx'+dx$ and $dy'+dx$, the equations

$$\sqrt{dx'^2+dy'^2} = \sqrt{dx^2+dy^2}, \qquad \tan^{-1}\frac{dy'}{dx'} = \tan^{-1}\frac{dy}{dx}$$

will in general have values depending upon that of the arbitrary ratio $dy : dx$. But they may be independent of this ratio: viz. this is the case when x', y' are functions of x, y such that

$$\frac{dx'}{dy} = -\frac{dy'}{dx}, \qquad \frac{dx'}{dx} = +\frac{dy'}{dy}.$$

and the two figures are then conformable (or conjugate) figures; that is, figures similar as regards corresponding infinitesimal elements of area. We have, in this case,

$$\sqrt{dx'^2 + dy'^2} = \sqrt{dx^2 + dy^2}, \qquad \tan^{-1}\frac{dy'}{dx'} - \tan^{-1}\frac{dy}{dx},$$

each a determinate function of x, y, the coordinates of P; and we pass from the element PQ to the corresponding element $P'Q'$ by altering the length in the ratio $\sqrt{dx'^2 + dy'^2} : \sqrt{dx^2 + dy^2}$, and rotating the element through the angle $\tan^{-1}\frac{dy'}{dx'} - \tan^{-1}\frac{dy}{dx}$; say, this ratio and this angle are the Aconic and the Strehlonic respectively, these being, as already mentioned, functions of x, y only.

Considering now any two conformable figures, say the figure of the points P, and that of the points P'; we have the theorem that we can from the first figure obtain a third conformable figure by means of an Aconic and a Strehlonic which are respectively equal to the Extension and the Rotation by which the second figure is derived from the first.

In fact, if in the three figures respectively we take x, y, x', y', and x'', y'', for the coordinates of the corresponding points P, P', P'', the first and second figures are conformable: and we have therefore

$$\frac{dx'}{dy} = -\frac{dy'}{dx}, \qquad \frac{dx'}{dx} = \frac{dy'}{dy};$$

the third figure is to have the Aconic $\sqrt{dx'^2 + dy'^2} : \sqrt{dx^2 + dy^2}$, and the Strehlonic

$$\tan^{-1}\frac{dy'}{dx'} - \tan^{-1}\frac{dy}{dx},$$

viz. writing r for $\sqrt{dx'^2 + dy'^2}$, we ought to have

$$dx'' = \frac{dx' + dy' \cdot dy}{r} dx, \qquad dy'' = \frac{-dx' \cdot dy + dy' \cdot dx}{r} dx,$$

and it is therefore to be shown that there exist x'', y'' functions of x, y satisfying these relations; for, this being so, we have

$$\frac{dx''}{dy} = -\frac{dy''}{dx}, \qquad \frac{dx''}{dx} = \frac{dy''}{dy};$$

and the third figure is then conformable with the first.

Writing, for shortness,

$$A = \frac{x'dx + y'dy}{r^2}, \qquad B = \frac{-x'dy + y'dx}{r^2},$$

the equations are

$$dx'' = A\,dx - B\,dy, \qquad dy'' = B\,dx + A\,dy,$$

or the conditions for the existence of the functions x'', y'' are

$$\frac{dA}{dx} + \frac{dB}{dy} = 0, \qquad \frac{dA}{dy} - \frac{dB}{dx} = 0.$$

We, in fact, have

$$\frac{dA}{dx} + \frac{dB}{dy} = \frac{1}{r^2}\left[\left(\frac{dx'}{dx} + \frac{dy'}{dy}\right)dx + \left(\frac{dy'}{dx} - \frac{dx'}{dy}\right)dy\right] + 2y\left\{-\frac{2}{r^4}\left[(xx'+yy')dx + (xy'-x'y)dy\right]\right\},$$

$$= \frac{2y}{r^2} - \frac{2}{r^4}(x^2+y^2)y,$$

and similarly

$$\frac{dA}{dy} - \frac{dB}{dx} = \frac{1}{r^2}\left[\left(\frac{dx'}{dy} + \frac{dy'}{dx}\right)dx + \left(\frac{dy'}{dy} - \frac{dx'}{dx}\right)dy\right] + 2x\left\{-\frac{2}{r^4}\left[(xx'+yy')dy - (xy'-x'y)dx\right]\right\},$$

$$= \frac{2x}{r^2} - \frac{2}{r^4}(x^2+y^2)x,$$

which proves the theorem.

The theorem is closely connected with the theory of the function of an imaginary variable; for, writing the problem for the conformable figures in the form

$$\frac{dx'}{dx} = \frac{dy'}{dy} = F, \qquad \frac{dx'}{dy} = -\frac{dy'}{dx} = G,$$

we have

$$dx' = F\,dx - G\,dy, \qquad dy' = G\,dx + F\,dy,$$

that is,

$$dx' + i\,dy' = (F+iG)(dx + i\,dy);$$

whence $F + iG$ is a function of $x + iy$, and thus by integration $x' + iy'$ is also a function of $x + iy$. In one point of view, any function such as $\phi(x,\,y) + i\,\psi(x,\,y)$ is a function of $x+iy$, for the quantity $x+iy$ is only known by means of its real components x, y; that is, knowing $x+iy$, we know x, y, and therefore the real function $\phi(x,\,y) + i\,\psi(x,\,y)$.

and Cauchy, adopting this definition, introduced the expression "function monogène" of $x + iy$, to denote that which is in the more restricted (and the ordinary) sense termed a function of $x + iy$. And MM. Briot and Bouquet, in their "Théorie des fonctions elliptiques" (Paris, 1875), although not using Cauchy's expression *fonction monogène*, but the simple term *fonction*, do this under the qualification stated p. 8: "*Pour ces fonctions* (or qui sont, sous un autre point de vue, des fonctions qui admettent *une dérivée*)." Now, a function admitting of a derivative (that is, in the ordinary sense, a function) of the imaginary variable $x + iy$, is a function such that, for a commutative value $x' = x + iy + dx + i\,dy$, we have a function $f(x') = f(x + iy)$, the condition is rather that $x' + iy'$ may be a function of $x + iy$;

=a quantity independent of the ratio of the real components dx, dy of the increment $dx + i\,dy$ of the imaginary variable. Or, what is the same thing, writing $f(x) = x' + iy'$, the condition is rather that $x' + iy'$ may be a function of $x + iy$;

$$dx' + i\,dy' = (F + iG)(dx + i\,dy);$$

where F and G are functions of x and y. It is not part of the condition that $F + iG$ shall be a function of $x + iy$; nor is it only a long way further on that the authors prove that this is the case (see the definition of a "fonction holomorphe," p. 14; and the proof, p. 137). The just-mentioned equation

$$dx' + i\,dy' = (F + iG)(dx + i\,dy),$$

where F and G are only assumed to be functions of x and y, has of course in the present paper not been used to define x', y' as conformable figures of x, y; for by means of the point P' with coordinates x', y', the geometrical interpretation that the figures of the points P and P' are conformable figures, that is, figures similar as regards their infinitesimal elements. The foregoing theorem in regard to the Aconics and the Strehlonics is that we can, by means of F and G, construct a third conformable figure, viz. in fact, knowing $x + iy$ as a function of $F + iG$ and $\tan^{-1}\dfrac{G}{F}$ respectively; and, using these as an Extension and a Rotation, we have the third conformable figure $x'' + iy'' = (F + iG)(x + iy)$; that is, $(F + iG)(x + iy)$, and therefore also $F + iG$, is a function of $x + iy$—and we have thus the derivative of a function of $x + iy$ as itself a function of $x + iy$.

It is to be remarked that, although the theorem of the Aconics and the Strehlonics, considered as a property of conformable figures, is now by any means geometrically self-evident, yet the foregoing analytical proof is only a proof conducted by means of real quantities; of when (admitting the theory of imaginary quantities) is in fact self-evident, viz. the analytical conclusion really is that, $F + iG$ denoting any function x, y, then, if $dx' + i\,dy' = (F + iG)(dx + i\,dy)$, that is, if $(F + iG)(dx + i\,dy)$ be a complete differential, then $F + iG$ is a function of $x + iy$.

730.

[ADDITION TO MR SPOTTISWOODE'S PAPER "ON THE TWENTY-ONE COORDINATES OF A CONIC IN SPACE."]

[From the *Proceedings of the London Mathematical Society*, vol. x. (1879), pp. 194–196.]

. WRITE

$$\begin{aligned} U &= (a,\, b,\, c,\, d,\, f,\, g,\, h,\, l,\, m,\, n \mid x,\, y,\, z,\, t)^2, \\ U'_x &= (\quad \mid \xi,\, \eta,\, \zeta,\, \omega)^2, \\ W &= (\quad \mid \xi,\, \eta,\, \zeta,\, \omega)^2, \\ P &= (a,\, b,\, c,\, d,\, f,\, g,\, h,\, l,\, m,\, n \mid \xi,\, \eta,\, \zeta,\, \omega \mid x,\, y,\, z,\, t), \\ P'_x &= (a,\, b,\, c,\, d,\, f,\, g,\, h,\, l,\, m,\, n \mid \xi,\, \eta,\, \zeta,\, \omega). \end{aligned}$$

Then the equation of the cone, having for its vertex the arbitrary point $(\xi,\, \eta,\, \zeta,\, \omega)$, and passing through the conic $U = 0$, $P = 0$, is

$$UP^2 - 2WP + UW = 0.$$

Or, if, to put the coefficients ξ, η, ζ, ω in evidence, we write for a moment

$$\begin{aligned} A &= (a,\, b,\, c,\, d \mid \xi,\, \eta,\, \zeta,\, \omega), \\ B &= (\quad \mid \xi,\, \eta,\, \zeta,\, \omega), \\ C &= (\quad \mid \xi,\, \eta,\, \zeta,\, \omega), \\ D &= (\quad \mid \xi,\, \eta,\, \zeta,\, \omega), \end{aligned}$$

and further

$$W = A\xi + B\eta + C\zeta + D\omega;$$

then the equation is

$$U(A\xi + B\eta + C\zeta + D\omega) - 2P^2(A\xi + B\eta + C\zeta + D\omega \mid Ax + By + Cz + D\omega)$$

$$+P^2(a,\, b,\, c,\, d,\, f,\, g,\, h,\, l,\, m,\, n \mid \xi,\, \eta,\, \zeta,\, \omega)^2 = 0.$$

And if we expand first in ξ, η, ζ, ω, and then in x, y, z, t, the final result is

	x^2	y^2	z^2	t^2	yz	zx	xy	xt	yt	zt
$+\xi^2$		C	B	A					h	g
$+\eta^2$	C		A	B	h					f
$+\zeta^2$	B	A		C	g	f				
$+\omega^2$	A	B	C				l	m	n	
$+\eta\zeta$		$3F$	$3E$	$\bullet$	$-2F$	$-2D$	$2E$	$\bullet$	$2(Q-B)$	$\bullet$
$+\zeta\xi$	$3F$		$3D$	$\bullet$	$-2C$	$-2F$	$2D$	$\bullet$	$\bullet$	$2(Q-B)$
$+\xi\eta$	$3E$	$3D$		$\bullet$	$2C$	$-2A$	$-2E$	$2(Q-B)$	$\bullet$	$\bullet$
$+\xi\omega$	$2L$	$3N$	$3M$	$3A$	$-2L$	$-2N$	$2M$	$-2(L-r)$	$-2N$	$-2N$
$+\eta\omega$	$3N$	$2M$	$3L$	$3B$	$2N$	$-2L$	$-2M$	$-2N$	$-2(M-q)$	$-2L$
$+\zeta\omega$	$3M$	$3L$	$2N$	$3C$	$-2M$	$2L$	$-2N$	$-2M$	$-2L$	$-2(N-p)$

In particular, if $a=0$, $f=0$, and $w=0$, we have for the foregoing equation $X=0$; and the like for the equations $Y=0$, $Z=0$, and $W=0$ respectively.

Take a, b, c, d, e, g, h for the six coordinates of the line through the points

$$\begin{vmatrix} x_1,\ y_1,\ z_1,\ t_1 \\ \xi_1,\ \eta_1,\ \zeta_1,\ \omega_1 \end{vmatrix},$$

then in, write

$$\begin{aligned} a &= y\zeta - \eta z, & l &= xa - \xi a,\\ b &= zx - \xi z, & g &= yb - \eta b,\\ c &= xy - \xi y, & h &= zc - \zeta c, \end{aligned}$$

where, of course,

$$af + bg + ch = 0.$$

Then the foregoing equation of the cone is

$$Ax^2 + By^2 + Cz^2 + Ft^2 + Gz^2 + Hy^2$$

$$-2A^2c - 2B^2a = 3Cab + (B^2a + DA^2)b + 2Cd\,(b - 2Ad)$$

$$+2Fd^2 + 2Bgh + 2L^2h$$

$$-2Eef - 2Egh + 2Hbh + 2Hde = 0.$$

And this may be regarded as the equation of the cone in terms of the twenty-one coordinates of the conic, and of the six coordinates of an arbitrary line meeting the cone. It is, in fact, the general form of the equation given in the paper—Cayley, "On a new Analytical Representation of a Curve in Space," *Quart. Math. Jour.*, vol. III. (1860), [**284**, this Collection, vol. IV. p. 452].

731.

ON THE BINOMIAL EQUATION $x^p - 1 = 0$; TRISECTION AND QUARTISECTION.

[From the *Proceedings of the London Mathematical Society*, vol. xi. (1880), pp. 4–17.
Read November 13, 1879.]

. THE solution of the binomial equation $x^p - 1 = 0$, p a prime number, or, what is the same, of the equation

$$x^{p-1} + x^{p-2} + \cdots + x + 1 = 0,$$

depends upon the Jacobian function

$$F_n x = x + ax^n + a^2 x^{n^2} + \cdots + a^{p-2} x^{n^{p-2}},$$

where a is a prime root of p, n any root whatever of the equation $x^{p-1} - 1 = 0$. Taking a a factor of $p-1$, and f for the supplementary factor (that is, $p-1 = af$), then, if n be a root of a, that is, root of $x^a - 1 = 0$, we have

$$F\eta = X_0 + \eta X_1 + \cdots + \eta^{a-1} X_{a-1},$$

where X_0, X_1, …, X_{a-1} denote each of them a period of f terms, viz.

$$X_k = x^{n^k} + x^{n^{k+a}} + x^{n^{k+2a}} + \cdots + x^{n^{k+(f-1)a}}.$$

$X_k = -1$, $\eta^a \ldots \eta^{a(p-1)}$ X_k -1, $\xi^a \ldots \xi^{a(p-1)}$.
(mod p); the values being roots of $a^a = 1$, and so for the other functions).

We have, of course, $F(1) = X_0 + X_1 + \cdots + X_{a-1}$, the sum of all the roots $= -1$; and, further, the general property that there is (independently of the value of a chosen periods is expressible as a sum

$$a_1 X_1 + a_2 X_2 + \cdots + a_{a-1} X_{a-1},$$

with known coefficients

$$a_1,\ a_2,\ \ldots,\ a_{a-1}.$$

The several cases $a = 2, 3, 4, \ldots$ may be termed those of the bisection, trisection, quartisection, &c., of the equation; viz.

$a = 2$, there are two periods, X, Y, and $F(= 1 + x + X + Y) = -1$;

$a = 3$, three periods, X, Y, Z, and $F\eta = X + \eta Y + \eta^2 Z$, η a root of $x^3 - 1 = 0$;

$a = 4$, four periods, X, Y, Z, W, and $F\theta = X + \theta Y + \theta^2 Z + \theta^3 W$, if θ be a root of $x^4 - 1 = 0$.

It is sufficient to attend to the prime roots which lie within the period p, and call these roots, $\eta_1, \eta_2, \ldots, \eta_{p-1}$ in respectively; and the relation between them is determined respectively, viz. if η be a b-th, we have always $F(\eta) = -1$; and if b be the of the periods for the bisection. The prime roots b are of course 1 and -1; and we have

$$F(1) = X + Y - 1 = -1,$$
$$F(-1) = X - Y = \sqrt{\pm p},$$

respectively.

As regards the bisection, it is known that $(X - Y)^2 = (-)^{(p-1)/2} p$, which is $+p$ or $-p$, according as p is $\equiv 1$ or $3 \pmod 4$; and the values of X, Y are thus determined. In what follows, I assume the case $p = 1 \pmod 4$, for which X, Y are thus determined, X, Y are the values $\frac{1}{2}(-1 \pm \sqrt{p})$.

It is to be remembered that, not the division into periods, but the order of the periods, depends on the choice of g, a prime root or pseudoprime of p; and, in what follows, I submit the prime root used to determine *Jacobi's* (J)*ables canoniques Primitifves suivis can Wurzeln der Einheit gebildet sind* (8vo, Berlin, 1872); and I have accordingly herein taken these as $\frac{1}{2}$-roots of Reuschle's selected primes.

For instance, $p = 13$, the powers of g are 1, 2, 4, 8, 3, 6, 12, 11, 9, 5, 10, 7; and, by writing these down in order in columns of 3 or 4, viz.

1	8	12
2	3	11
4	6	9

1	8	12	5
2	3	11	10
4	6	9	7

we have the periods X, Y, Z as X, Y, Z, W, belonging to the trisection and the quartisection of $p = 13$.

I further remark that the equations which I am concerned with are all given in Reuschle, but in a somewhat different form; thus, $p = 13$, quartisection (see p. 10), he has

$$\eta^4 = -\eta^3 - 2\eta + 3, \qquad \eta_1\eta_2 = -3\eta_1 - \eta_2 + 3, \quad \eta_1\eta_3 = -1 - \eta_3,$$

(where observe that here and in every case the value of $\eta_1\eta_3$ is at once obtained from that of $\eta_1\eta_2$ by mere cyclical interchange of the suffixes, so that the last equation is in fact superfluous), the other equations, using $\eta_1+\eta_2+\eta_3+\eta_4 = -1$, can be eliminated; the result is system of equations, $X+YZ = 0$, $XY+Z = 0$, &c.

Similarly, in the case of a trisection, the equation for $\eta_1\eta_2$ is superfluous, and the other equations give my values of X^2 and XY.

Reuschle gives also, and I take from him, the cubic and the quartic equations satisfied by η, $= 1$, b, ..., n for the various periods of those obtained for X, Y, Z, ... and connections with these periods is at once seen the rule of formation of those periods in the trisections and the quartisections respectively.

Many of the results obtained accord with, and furnish exemplifications of general theorems contained in Jacobi's memoir, "Ueber die Kreistheilung und ihre Anwendung auf die Zahlentheorie," *Crelle*, t. XXX. (1846), pp. 166–182; [*Ges. Werke*, t. VI. pp. 254–274].

Trisection, $a = 3$, $p \equiv 1 \pmod 3$**.**

We have there periods X, Y, Z, and we choose

$$XY = \tfrac{1}{3}(p-1) + aX + bY + cZ,$$

$$XY = (f,\ g,\ h,\ k)$$

the coefficients a, b, c, f, g, h being determinate integers. And, by cyclical interchanges, we obtain equations which may be written

$$\begin{aligned} X^2 &= a,\ b,\ c, \\ Y^2 &= b,\ c,\ a, \\ Z^2 &= c,\ a,\ b, \\ XY &= f,\ g,\ h, \\ YZ &= g,\ h,\ f, \\ ZX &= h,\ f,\ g, \end{aligned}$$

viz. here and elsewhere the coefficients a, b, c are written to denote the sum

$$aX + bY + cZ.$$

It is easy to see that

$$f + g + h = (p-1).$$

in fact, multiplying $X^2+Y^2+Z^2$ together, the coefficient of X is $a+c+b$, and there are no terms the product of which is unity; hence XY contains $\frac{1}{3}(p-1)$ unit terms, each a power of g, and this sum is XY.

From the equation $X+Y+Z=-1$, multiplying by X, and for XY, XZ substituting their values, we obtain an expression

$$(a+f+h)X+(b+f+g)Y+(c+g+h)Z=-\tfrac{1}{3}(p-1)X,$$

which must identically vanish; viz. the three coefficients must be each of them $=0$; we must have

$$\begin{aligned} a+f+h&=-p,\\ b+f+g&=0,\\ c+g+h&=0, \end{aligned}$$

so that, taking f, g, h as known, the other coefficients a, b, c are given in terms of them. The equations give

$$\begin{aligned} X^2&=YZ+X(\tfrac{1}{3}p+g+h),\\ Y^2&=ZX+Y(\tfrac{1}{3}p+h+f),\\ Z^2&=XY+Z(\tfrac{1}{3}p+f+g). \end{aligned}$$

We have X, $YZ=Y$, $XZ=Z$, XY, $(p-1)$, so that, substituting for X^2, XY, etc., their values,

$$ah+bk+ck=-p(a+b+c)+4(f+g+h)k,$$

$$\begin{aligned} af(\eta+h)+4+h(\eta+f)&=(4+h+f)+4+f(\eta+g),\\ ag(g+f)+g(g+f)&=(\eta+g+f)k+g(\eta+g), \end{aligned}$$

that is,

$$\begin{aligned} ah+bk+ck&=pf+pg+2k(k+l),\\ bh+g(g+h)&=ah+g(h+l)=ah+g(b+l),\\ gh+g(g+l)&=(g+l+h)k+g(g+l), \end{aligned}$$

equations which reduce themselves to the single equation

$$gh+h^2+g^2=-\tfrac{1}{3}(p-1)-\tfrac{1}{3}(g+h)-\tfrac{1}{3}-\tfrac{1}{3}(g-h).$$

and this is the only relation obtainable by consideration of the three equal values

$$X,\, YZ, \quad Y,\, ZX, \quad Z,\, XY.$$

Moreover, this equation being satisfied, the six functions in the three equations because each of them $=fg-fg$; so we have, from the three equations becomes

$$XYZ=(fg-fg)/p-h^2,$$

that is,

$$XYZ=fp-h^2.$$

We have

$$\begin{aligned} F_p\cdot F_p^2&=X^2+Y^2Z+\eta YZ\cdot YZ-X\cdot XY\\ &=(\eta+\eta+\eta)-b\eta-c)(X+Y+Z)\\ &=-\eta+\eta+\eta-b\eta-c\\ &=3(f+g+h) \end{aligned}$$

that is,

$$F_\eta F_\eta^2 = p.$$

We have, moreover,

$$(F_\eta)^3 = X^3 + Y^3 + Z^3 + \eta \cdot 3X^2Y + \eta^2 \cdot 3XY^2 + (3X^2Z + \eta + \eta^2)$$

$$= [\eta\eta'\eta'',\ a + b\eta + \eta'\eta'(X - Y) + (Y - Z) + (Z - X)\, XY\eta\eta',\ +3XYZ];$$

which is

$$= \tfrac{1}{2}(a + 4b - 4c + (-3a + 4b + c)\eta + (3a + b + 4c)\eta^2),$$

as is at once seen by comparing the coefficients of X, Y, Z respectively.

Hence, writing

$$a + b(\eta + \eta'') + c(\eta'' + \eta) = a^2 + b^2 + c^2 - bc(\eta - 1) - (-1)$$

$$= a + 2b - c,$$

we have

$$A = a + 2b - c, \quad B = a + 2c - b, \quad C = b - c;$$

η a root of $\eta^2 + \eta + 1 = 0$.

We have also,

$$(F_\eta)^3 = (a + b\eta + c\eta'')X + (a + b\eta'' + c\eta)Y + (a + b\eta + c\eta'')Z,$$

and these together

$$= (Ap + Bq)/p,$$

equations which reduce themselves to the single equation

$$gh + h^2 + g^2 = -\tfrac{1}{3}(p - 1)$$

or, say

$$(F_\eta)^3 = p\,(F_\eta)/F_\eta;$$

viz. here it has, X, $YZ = Y$, $XZ = Z$, XY, $(p - 1)$, so that, substituting for X^2, XY, etc., the equation becomes

$$XYZ = (Ap + Bq)/p - h^2,$$

or the expression $A + Bi$ determined as above is a complex factor of p.

As already remarked, the values of a, b, c, f, g, h; and the equations in a, b in effect given in Reuschle; the complete factors of p, as given in p. 9, are $4p = a^2 + 27b^2$, where reduced to the form $A + Bi$, are not identical with the $A + Bi$ found from the foregoing theory.
A and B being respectively

$$a^2 - bc = \tfrac{1}{2}(p - 1) - (f - g) \cdot a^2 - bc(\eta - 1) - (-1)$$

$$= a + 2b - c.$$

And the A and B in Reuschle's selected primes; the for the primes 7, 13, …, 97, the values from Reuschle of a, b, c, f, g, h, and of the coefficients of the η-equation; also the values of A and B derived from f, g, h by the foregoing formulæ. It will be seen that all the values are consistent with the theory.

TABLE FOR THE TRISECTION.

p	a, f	b, g	c, h	a^2+27b^2	A	B	Page in Reuschle	
7	−2, 1	−1, 1	−1, 2	−2, 1	1	2	1	p. 5
13	−4, 3	−3, 2	0, 4	−4, 3	−4	−3	0	p. 15
19	−4, 3	−1, 3	1, 7	−6, 7	−7	−3	3	p. 16
31	−7, 5	−1, 5	1, 7	−10, 1	−2	5	−2	p. 18
37	−8, 4	−2, 4	4, 3	−12, 11	−4	3	−5	p. 34
43	−4, 3	−10, 7	−7, 4	−12, 15	−4	3	0	p. 69
61	−11, 9	−12, 11	−11, 11	−20, 33	−9	8	−1	p. 97
67	−16, 11	−2, 3	−7, 8	−22, 17	5	2	9	p. 107
73	−16, 11	−12, 7	−4, 2	−24, 27	−7	−9	9	p. 128
79	−20, 17	−16, 11	−26, 21	−26, 33	−10	−3	9	p. 154
97	−29, 21	−26, 21	−36, 29	−32, 79	11	9	4	p. 187

Quartisection, $a = 4$, $p \equiv 1 \pmod 4$**.**

We have four periods X, Y, Z, W; and we obtain

$$\begin{aligned} X^2 &= (a,\, b,\, c,\, d \mid X,\, Y,\, Z,\, W), \\ XY^2 &= (f,\, g,\, h,\, k), \\ XZ &= (l,\, m,\, n,\, p). \end{aligned}$$

the coefficients being determinate integers. It can be shown that $l = m = \frac{1}{4}(p-1)$, and $= \frac{1}{4}(p+1)$ according as $p \equiv 1$ or $5 \pmod 8$. And then, by cyclical interchanges,

$$\begin{aligned} X^2 &= a,\, b,\, c,\, d, \\ Y^2 &= b,\, c,\, d,\, a, \\ Z^2 &= c,\, d,\, a,\, b, \\ W^2 &= d,\, a,\, b,\, c, \\ XY &= f,\, g,\, h,\, k, \\ YZ &= g,\, h,\, k,\, f, \\ ZW &= h,\, k,\, f,\, g, \\ WX &= k,\, f,\, g,\, h, \\ XZ &= l,\, m,\, n,\, p, \\ YW &= m,\, n,\, p,\, l. \end{aligned}$$

We have, as in the manner as for the trisection,

$$f + g + h + k = \tfrac{1}{4}(p-1),$$

and so also the equation

$$2(X + Y + Z + W) = XY + YZ + ZW + WX + XZ + YW + ZW,$$

and, in virtue of the foregoing value of $l + m = -\frac{1}{4}(p-1)$ or $\frac{1}{4}(p-5)$ according as $p \equiv 1$ or $5 \pmod 8$.

Again, from the equation $X + Y + Z + W = -1$, multiplying by X and reducing, and thence

$$\begin{aligned} a + b + c + d &= -2(f + g + h + k) - 2(l + m), \\ &= -\tfrac{1}{2}(p-1) - 2(l + m). \end{aligned}$$

We have

$$XYZ = YXZ = ZXY,$$

that is,

$$X(g,\, h,\, k,\, f) = Y(l,\, m,\, n,\, p) = Z(f,\, g,\, h,\, k),$$

and thence

$$\begin{aligned} g(a,\, b,\, c,\, d) + h(f,\, g,\, h,\, k) + k(g,\, h,\, k,\, f) + f(l,\, m,\, n,\, p) = \\ l(b,\, c,\, d,\, a) + m(g,\, h,\, k,\, f) + n(c,\, d,\, a,\, b) + p(h,\, k,\, f,\, g) = \\ f(c,\, d,\, a,\, b) + g(h,\, k,\, f,\, g) + h(l,\, m,\, n,\, p) + k(d,\, a,\, b,\, c) = \end{aligned}$$

that is,

$$\begin{aligned} ka + ga + h^2 + k^2 + gh + (f+l)a + (g+p)b + (l+g)c + (k+f)d = 0, \\ hb + ka + h^2 + kf + gh + (f+l)a + (h+l)b + (m+n)c + (k+l)d = 0, \\ lc + ld + hf + kg + (h+l+g)a + (g+l)c + gh + (l+g)c + (m+l)d = 0, \\ ld + ld + hf + kg + (g+l+h)a + g(g+l) + (k+l)d + g(g+l) = 0, \end{aligned}$$

in which equations a, b, c, d may be regarded as having their foregoing values.

One of these equations is

$$hc + fa + gd + bd + (a+g)c + (f+p)d = bf,$$

that is,

$$-b + h + a + l + b(a+g) + p(g+p) + cl = -p(f+p) + p(g+p) + d,$$

or, reducing,

$$l(g+d-1) - f - p = (a+b)c - 2bp + g(d+b) - p(g+p) + dp,$$

which gives

$$l(g+d-1) - f - p = (a+b)c - 2bp + g(d+b) - p(g+p) + dp - p^2,$$

Again, another equation is

$$hb + g(g+l) + p(g+p) + bp = lp,$$

that is,

$$-h(g+l+n) + l(l+p) + dl + b(l+p) = a(l+p) + lp + bp,$$

or, reducing,

$$m(g+k-l) = ld + h(l+p) + d - (l+p),$$

which gives

$$m(g+k-l) = ld + h(l+p) + d - (l+p),$$

And we have also

$$md + l + n = a + ld + h + h(d+p) - 2p,$$

that is,

$$= m(a+b+n) + h(g+b) + b(g+a) + p(g+p) + (b+p) + l,$$

or, reducing,

$$b(a+l+p) + b(a+p) + (a+p)l = (a+p)l + (a+p)b + (a+p) + p(g+p) + (a+p)d,$$

Substituting herein for b, c, their values, we have

$$(b+l)(2f+p-28f-lp+k(f+1)(2g-p-lp+h(g+1)(2f+p-lp+h(g+1)(2p-p-lp)+(k-p)(g+p+2(p-p))+(k-p)(g+p+2(p-p))$$

12—2

In this equation the only terms of the second order are $-(b+2p$, which contains the factor b; the terms of the third order contain this same factor b, and throwing it out, and reducing, the equation is found to be

$$(g-h)^2+4l=-f^2-h-k-p,$$

or, as it may also be written,

$$g^2+h^2-2gh-1+(\tfrac{1}{2}h^2-1+2-2p-h-kp)=0,$$

and the foregoing values of l, m are

$$\frac{1}{2}(g+h-2k+2-2p)(f-g)-(gh-lp)= \ -m=\frac{1}{2}(g+h-2k+2-2p)(f-g)-(gh-lp)=$$

and by means of these equations all the foregoing equations are satisfied.

We have

$$F(\eta)=(X-Y)+i(\eta(Y-W)),$$
$$=X-Y+i(\eta(Y-W))-i(2(XY+YZ)-YW),$$
$$=(-1+a+b+a)+i(\eta b),$$

or, substituting for a, b, c, d, and for f, g, h, k in evidence, we write

$$F(\eta)=(\alpha)^{p-1}\,p.$$

Again, we have

$$(F(\eta))^2=(X-W)^2+2(\eta(Y-W)),$$
$$=X^2-Y^2-Z^2-W^2+2(XY-YZ+ZW+XY)-WX)+2i(2XY-ZW-WX)+WX,$$
$$=(a-b-c-d)+2(f+g-h+k)-l+m)-2(2(p+2h+2(g-h)-2(g+l)l)),$$
$$=(a-b)X^2+2(2k-2l+h(g-h))(g-h)),$$

where

$$A=a+b-c+d-(b-c)+d+2(f-g-h+k)-2(l-m),$$
$$B=2(f-g+h-k);$$

or, that $X-Y-W=(a+p\,F(\eta))-g+(a+h(g-h))$.

The function A^2+Bi is here a complex factor of p, viz. we have

$$F(\eta)F'(\eta)=(A+B)^2(A-B^2)=A^2+B^2,$$

and similarly

$$(F'(\eta))F(\eta)=(A-B)^2(F(\eta))-1.$$

Moreover
$$F'(-1)(-1) = (-)^{(p-1)/4}\,p.$$
and we have therefore
$$(F'(-1))^2 = (A^2+B^2)\,p,$$
that is,
$$A^2+B^2 = p,$$
or the expression $A+Bi$ determined as above is a complex factor of p.

Taking the first expression of those above written for A, B, we have

$$a+b+c+d = -\tfrac{1}{2}(p-1) - 2(l+m),$$

$$\begin{aligned}+A &= a-b+c-d,\\ &= 2a+2c+\tfrac{1}{2}(p-1)+2(l+m),\\ &= 2(a+c+l+m)+\tfrac{1}{2}(p-1).\end{aligned}$$

We had A, YZ writes equally as the product of XY and of XW, viz. we obtain

$$XYZW = a + b(g,\,h,\,k,\,f) + c(b,\,c,\,d,\,a) + d(g,\,h,\,k,\,f)$$

$$\begin{aligned}&= a+b+b-h-k-f-(f+l)a+(g+p)b+(l+g)c+(k+f)d,\\ &= 2(f+g)h(a+b+d)+2(f+g)l = 2g(g+p)k-2g(a+p)d-g\eta+h(b+c+d),\\ &= 2(f+g)l(b+d)-p(g+p)+(g+p)d.\end{aligned}$$

We had $XYZW$ unable as the product of XY and of ZW, viz. we obtain

$$XYZW = (f,\,g,\,h,\,k)(b,\,c,\,d,\,a)$$

$$\begin{aligned}&= f(b+l)+d(b+l)(b+d)-(b+l+d)(f+l)+l(b+g)d,\\ &= (f+l)k-g-2bl+(a+b)-(b+l+d)l+d(b+l+d)+lp,\\ &= (f+l)(k-l+p)+d(b+p)l-(b+l+d)l+d(b+l+d)-lp,\\ &= (f+l)(k-p)+d(g-l+p)-p(l+b+p)l,\end{aligned}$$

that is,
$$XYZW = (f+l)(k-p) - bl + d(b+p)\,d + 2,(p+l+p)(b+d),$$
where, for the sake of having a single formula, I have retained l in place of the value $-\tfrac{1}{2}(p-1+l)-\tfrac{1}{2}(p-8)$ according as $p\equiv 1$ or $5 \pmod 8$.

We then have the following.

TABLE FOR THE QUARTISECTION.

p	a	b	c	d	f	g	h	k	l	m	A	B	Page in Reuschle
5	0	1	1	0	0	1	1	0	−1	−1	1	−2	p. 4
13	0	1	0	0	1	0	1	1	−1	0	2	−3	p. 5
17	4	−2	−2	0	0	1	2	1	−1	−1	−2	2	p. 12
29	4	4	−3	−2	1	4	4	1	−1	−2	−7	−1	p. 33
37	0	5	4	4	2	4	5	2	−2	−3	−3	4	p. 84
41	−10	−5	−2	5	4	1	1	4	−2	−3	−1	−2	p. 86
53	0	7	4	4	5	4	7	5	−2	−2	5	5	p. 99
61	4	4	−7	−12	5	7	7	5	−1	−3	−5	−7	p. 106
73	−10	−16	−12	0	7	4	5	7	−3	−2	7	−4	p. 120
89	−20	−19	−13	−11	7	5	5	7	−3	−4	−5	−8	p. 151
97	−22	−16	−17	−19	10	5	5	5	−3	−4	9	−4	p. 187

[Figure: large table of computed values, transcribed above.]

Table of the Powers of Reuschle's Selected Prime Roots.

[Figure: a large arithmetical table 65 columns wide listing powers $g^k \pmod p$ for selected primes $p = 5, 7, 11, 13, 17, 19, 23, 29, 31, 37, 41, 43, 47, 53, 59, 61, 67, 71, 73, 79, 83, 89, 97$ across the columns, and exponents $k = 1, 2, 3, \ldots$ down the rows. The cells of this table contain entries such as $1, 2, 4, 8, 16, \ldots$ for $p = 17, g = 3$, etc. The full table fills the page.]

Table (continued).

[Figure: continuation of the table of powers of Reuschle's selected prime roots, spanning further values of the exponent k for the primes listed on the preceding page. Numerous numerical entries fill the page, in the same format as the previous table.]

732.

A THEOREM IN SPHERICAL TRIGONOMETRY.

[From the *Proceedings of the London Mathematical Society*, vol. xi. (1880), pp. 48–50.
Read January 8, 1880.]

. In a spherical triangle, where a, b, c are the sides, and A, B, C their opposite angles, we have

$$\tan\tfrac{1}{2}a\ \tan\tfrac{1}{2}c = \tan\tfrac{1}{2}b\sin(A-B) + \tan\tfrac{1}{2}b\sin A - \tan\tfrac{1}{2}a\sin B,$$
$$\tan\tfrac{1}{2}c\,(1-\tan\tfrac{1}{2}a\ \tan\tfrac{1}{2}b\ \cos(A-B)) = \tan\tfrac{1}{2}b\ \cos A + \tan\tfrac{1}{2}a\ \cos B,$$

which are both included in the form

$$\tan\tfrac{1}{2}c\cos B = \frac{\tan\tfrac{1}{2}b\sin(A-B)\,(\cos A + i\sin A)}{1+\tan\tfrac{1}{2}a\ \tan\tfrac{1}{2}b\,(\cos A + i\sin A)}.$$

For the first of the two identities: from

$$\cos a = \frac{\cos A + \cos B\ \cos C}{\sin B\ \sin C},$$
$$\cos b = \frac{\cos B + \cos C\ \cos A}{\sin C\ \sin A},$$

we deduce

$$\begin{aligned}
\cos a - \cos b &= \frac{1}{\sin A}\Big(\frac{\cos A}{\sin C\ \cos B} - \frac{\cos B}{\sin C\ \cos A}\Big) - \frac{1}{\sin B}\Big(\frac{\cos B}{\sin C\ \cos A} - \frac{\cos A}{\sin C\ \cos B}\Big),\\
&= \frac{1}{\sin A}\ \frac{(\sin A\sin B - \sin B\sin A)\,(\cos(A-B) - \cos A\sin B)}{\sin C\sin A\sin B},\\
&= \frac{\sin(A-B)}{\sin A\sin B\sin C}\ \big(\cos A\sin B - \cos B\sin A\big),\\
&= -\frac{\sin(A-B)}{\sin C}\ (\cos A\sin B - \cos B\sin A),\\
&= \frac{\sin(A-B)}{\sin C}(\cos(B+A) - 1),
\end{aligned}$$

that is,

$$-\sin(A-B) = \frac{\sin C}{\sin a \sin b}\bigl(\cos a \sin b - \cos b \sin a\bigr);$$

or, what is the same thing,

$$-\tan\tfrac{1}{2}a\tan\tfrac{1}{2}b\sin(A-B) = \frac{\sin C}{\sin a \sin b}(\cos a \sin b - \cos b \sin a),$$

Here $\cos a - \cos b = (1+\cos a) - (1+\cos b)$; substituting for $\dfrac{\sin a}{\sin b}$ correspondingly $\dfrac{\sin A}{\sin B}$ and $\dfrac{\sin B}{\sin A}$, the right-hand side is

$$= \frac{1+\cos a}{\sin a \sin A} - \frac{1+\cos b}{\sin b \sin B},$$

$$= \cot\tfrac{1}{2}a - \cot\tfrac{1}{2}b,$$

whence, multiplying each side by $\tan\frac{1}{2}a\ \tan\frac{1}{2}b$, we have the relation in question.

For the second identity which is

$$\tan\tfrac{1}{2}c\,[1-\tan\tfrac{1}{2}a\ \tan\tfrac{1}{2}b\ \cos(A-B)] = \tan\tfrac{1}{2}b\ \cos A + \tan\tfrac{1}{2}a\ \cos B;$$

if on the right-hand side we substitute for $\sin A$, $\sin B$ their values

$$\frac{\cos a + \cos b \cos c}{\sin a \sin b} \quad\text{and}\quad \frac{\cos b + \cos c \cos a}{\sin b \sin a},$$

the right-hand side becomes

$$\frac{1}{\sin C}\bigl[\cos a - \cos b + \cos C(\cos a \cos b - \cos b \cos a)\bigr],$$

whence, multiplying the whole equation by $\sin C$, it becomes

$$(1-\cos c)(1+\cos a)(1-\cos b) = \tan\tfrac{1}{2}b\cos(A-B)$$

$$= (1-\cos c)(\cos a - \cos b) + (1+\cos c)(\sin a \cos b - \cos a \sin b).$$

We have here

$$\cos(A-B) = \frac{\cos a \sin b \cos B + \sin a \cos B}{\sin a \sin b};$$

by substituting for $\sin a$ and $\sin b$ their values, and so for A, B their values

$$\frac{\cos b}{\sin c \cos a + \cos c} \quad \text{and} \quad \text{etc.} \quad \text{where}$$

$$\cos a + \cos b + \cos C = -\cos a + \cos b + \cos a \cos b + \cos C,$$

or

$$\square = 1 - \cos^2 a - \cos^2 b - \cos^2 c + 2\cos a \cos b \cos c.$$

The numerator is

$$\cos C \sin b - \cos\,(\cos^2 a + \cos^2 c) \sin a\; \sin b + \cos a \sin b$$

$$= 1 - \cos^2 c - (\cos^2 a + \cos^2 c) \sin a\; \sin b + \cos a \sin b + 2 \cos a;$$

viz. this is

$$= \cos a \sin b\,(1 + \cos a) - (\cos^2 a + \cos^2 c)(1 + \cos a)(1 - \cos a) - 1 - \cos a;$$

having the factor $1 + \cos a$, which is also a factor of $\sin^2 a, = 1 - \cos^2 a$, in the denominator. We have, therefore,

$$\cos(A - B) = \frac{\cos a \sin b \cos(1 + \cos a) - (\cos^2 a + \cos^2 c) \sin a\; \cos b - 1 - \cos a}{(1 - \cos a) \sin a \sin b},$$

and the equation thus is

$$(1 - \cos c)(1 + \cos a)(1 - \cos b) + \cos a \sin b\; \cos(1 + \cos a) - (\cos^2 a + \cos^2 c) \sin a\; \cos b - 1 - \cos a$$

$$= (1 - \cos c)(\cos a - \cos b) + (1 + \cos c)(\sin a \cos b - \cos a \sin b),$$

where each side is in fact

$$= \cos a \sin b\,(1 + \cos c) + \cos a \sin b\,(1 + \cos c) - \square = 1 - \cos c\; \sin b;$$

and the second identity is thus proved.

733.

ON A FORMULA OF ELIMINATION.

[From the *Proceedings of the London Mathematical Society*, vol. xi. (1880), pp. 139–141.
Read June 10, 1880.]

. CONSIDER the equations

$$(a, \ldots \mid \xi, \eta)^p = 0, \qquad (A, \ldots \mid \xi, \eta)^q = 0,$$

where $a, \ldots, A, \ldots$ are functions of coordinates. To fix the ideas, suppose that each of these coefficients is a linear function of the four coordinates x, y, z, w. Then, eliminating θ, we obtain $\nabla = 0$, the equation of a nodal-surface (the form of ∇ is known); this surface has a nodal curve.

It is easy to obtain the equations of the nodal curve in the case where one of the equations, say the second, is a quadric: the process is substantially the same whatever may be the order of the other equation, and I take it to be a cubic: the two equations therefore are

$$(a, b, c, d \mid \xi, \eta)^3 = 0,$$

$$(A, B, C \mid \xi, \eta)^2 = 0;$$

giving rise to an equation

$$\nabla = (a, b, c, A, B, C)^2 = 0.$$

And it is required to perform the elimination so as to put in evidence the nodal line of this surface.

Take θ_1, θ_2 the roots of the second equation, or write

$$(A, B, C \mid \theta, 1)^2 = A(\theta - \theta_1)(\theta - \theta_2);$$

that is,

$$\theta_1 + \theta_2 = -\frac{2B}{A}, \qquad \theta_1\theta_2 = \frac{C}{A}.$$

then, if

$$\Theta_1 = (a,\, b,\, c,\, d \mid \xi,\, \eta)^3,$$
$$\Theta_2 = (a,\, b,\, c,\, d \mid \xi,\, \eta)^3,$$

we have

$$\nabla = A^4\, \Theta_1\, \Theta_2.$$

viz. on the right-hand side, replacing the symmetrical functions of θ_1, θ_2 by their values in terms of A, B, C we have the expression of ∇ in the form

$$\nabla = a^2C^2 + etc.$$

Form now the expressions

$$\Theta_1 = \theta_1X_1 + \theta_1^2X_2 + \theta_1^3X_3, \quad \Theta_2 = \theta_2X_1 + \theta_2^2X_2 + \theta_2^3X_3,$$

each divisible by $\theta_1 - \theta_2$. These are evidently symmetrical functions of θ_1, θ_2, the values being given by the successive lines of the determinant

$$\begin{array}{llll} 0, & 1, & \theta_1+\theta_2, & \theta_1^2+\theta_1\theta_2+\theta_2^2 \quad (\text{\&c. etc.}), \\ -1, & 0, & \theta_1\theta_2, & \theta_1\theta_2(\theta_1+\theta_2), \\ -(\theta_1+\theta_2), & -\theta_1\theta_2, & 0, & \theta_1^2\theta_2^2, \\ -(\theta_1^2+\theta_1\theta_2+\theta_2^2), & -\theta_1\theta_2(\theta_1+\theta_2), & -\theta_1^2\theta_2^2, & 0, \end{array}$$

and, consequently, these same quantities, each multiplied by A^3, are given by the consecutive lines of

$$\begin{array}{cccc} 0, & A^2, & -2AB, & AC+4B^2 \quad (\text{\&c. etc.}), \\ -A^2, & 0, & AC, & C^2-2BC, \\ 2AB, & -AC, & 0, & C^2, \\ AC+4B^2, & 2BC, & -C^2, & 0. \end{array}$$

Calling these X, Y, Z, W, that is, writing

$$X = 3Av - 6Bb + (-AC + 4B^2)a, \quad \text{\&c.},$$

then X, Y, Z, W are the values of

$$\Theta_1 - \Theta_2, \quad \Theta_1\Theta_2, \quad \Theta_1^2\Theta_2, \quad \Theta_1\Theta_2^2,$$

each multiplied by $A^3 = (\theta_1 - \theta_2)$; and the functions all four of them vanish if only $\theta_1 = \theta_2$, or, what is the same thing, the equations $X = 0$, $Y = 0$, $Z = 0$, $W = 0$ constitute only a twofold system.

The functions

$$\big(X,\, Y,\, Z\big), \quad \big(Y,\, Z,\, W\big)$$